Digital Circuits

Exercise 1- Digital Building Block Measurements

Group/Team No.: Group 1____

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1st lab exercise	**max score**	**score reached**
Security training	mandatory	0
Preparation	0	0
Task 1 (static behavior)	2	
Task 2.1 (generator adjustment)	2	
Task 2.2 (rise/fall time, delay)	3	
Task 2.3 (delay)	1	
Task 3.1 (glitch)	3	
Task 4.1 (oscillator)	2	
Sum	**13**	
Protocol	**5**	
Penalty		
Total points	**18**	

1 Task 1.1, Behavior of a gate with static power supply

In this experiment, one 7400 NAND gate IC was placed on the evaluation board. The IC was powered with Vcc = 3.3 V and GND = 0 V (pin 14 = Vcc, pin 7 = GND). One NAND gate from the chip was connected so that the two inputs were controlled by switches, and the output was connected to an LED through a resistor. The setup was used to test the logic behavior of the NAND gate. Each possible input combination (A, B) was applied and the LED indicated the output. The observed truth table matched the expected behavior of a NAND gate: 0 0 1
0 1 1
1 0 1
1 1 0

2 Task 2 Timing characteristics of a gate

2.1 Function Generator Adjustment

Task 2.1 – Function Generator Adjustment and Observation In this task, the function generator output was connected directly to oscilloscope channel 1 using a terminated coaxial cable. No digital circuit was used in this step. The function generator was adjusted to the following parameters: Frequency: between 10 kHz and 100 kHz Amplitude: 3.3 V peak-to-peak (Vpp) DC Offset: adjusted so that the signal alternates between 0 V and 3.3 V During the experiment, a square wave signal was observed on the oscilloscope, alternating between 0 V and 3.3 V. This was the expected behavior, as the square wave represents a digital "0" and "1," confirming that the function generator was configured correctly.

2.2 Rise/Fall Time and Propagation Delay We measured how fast a NAND gate responds to input changes:

- Input rise time (time for input to go from low to high)
- Output fall time (time for output to go from high to low)
- Propagation delay (time difference between input and output changes)

Experimental Setup:

- Function generator connected to NAND gate input (CH1)
- NAND gate output measured on CH2
- Oscilloscope used to observe signals:
  - CH1 (input): 2 V/div
  - CH2 (output): 10 V/div

Observations:

- Input voltage rises from 0 V to 3.3 V
- Output voltage falls from 3.3 V to 0 V (NAND behavior)
- Small overshoot (<10%) visible on output due to cables/probes
- Signals are smooth and clean

Measurements and Calculations

1. Input Rise Time (t_r)

Rise time = time for voltage to go from 10% to 90% of full swing.
Input voltage: 0 V → 3.3 V

- 10% = 0.33 V, 90% = 2.97 V
- Input edge spans about 0.5 divisions horizontally

Formula:

$$t_r = (number of divisions) \times (time per division)$$

Calculation:

$$t_r = 0.5 \times 5\ ns = 2.5\ ns$$

2. Output Fall Time (t_f)

Fall time = time for voltage to go from 90% to 10% of full swing.
Output voltage: 3.3 V → 0 V

- 90% = 2.97 V, 10% = 0.33 V
- Edge spans about 0.7 divisions horizontally

Formula:

$$t_f = (number of divisions) \times (time per division)$$

Calculation:

$$t_f = 0.7 \times 5\ ns = 3.5\ ns$$

3. Propagation Delay (t_{pd})

Propagation delay = time difference between the 50% point of input and output.
Input 50% = 1.65 V, Output 50% = 1.65 V

- Horizontal distance ≈ 1 division

Formula:

$$t_{pd} = (horizontal divisions between 50\% points) \times (time per division)$$

Calculation:

$$t_{pd} = 1 \times 5\ ns = 5\ ns$$

Results Summary

cc

Parameter	Value	Comment
Input rise time	2.5 ns	From 0 → 3.3 V (10% → 90%)
Output fall time	3.5 ns	From 3.3 → 0 V (90% → 10%)
Propagation delay	5 ns	Time from input 50% → output 50%
Overshoot	<10%	Caused by cables/probes

- Input rises faster than output falls because the gate takes time to drive the output.
- Propagation delay (∼5 ns) is normal for a CMOS NAND gate.
- Small overshoot is due to cables/probes, not the gate itself. .

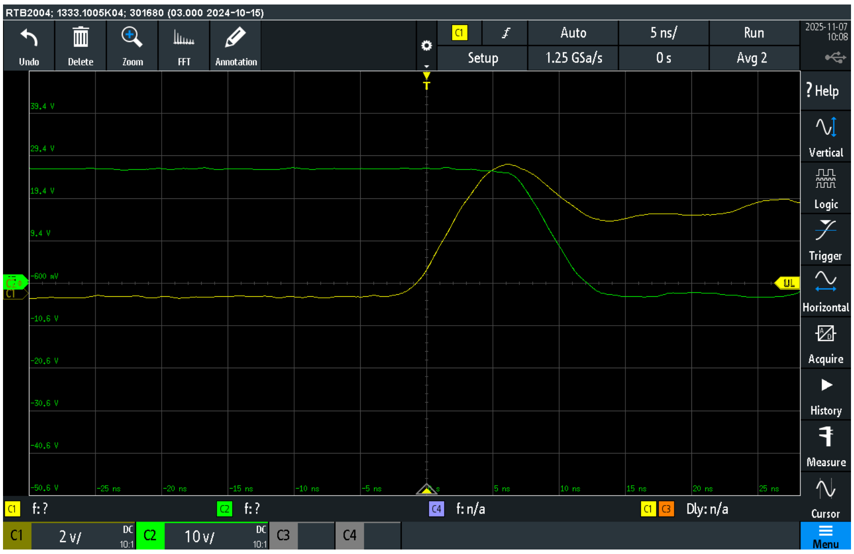

Figure 1: Input and output waveforms of the NAND gate

Task 2.3 Propagation Delay Measurement

In this task we measured the propagation delay. The input signal was applied to channel C1 (yellow) and the output signal was observed on channel C2 (green) using the oscilloscope. The probe was set to 10:1, vertical scale 2 V/div, horizontal scale 10 ns/div, and sample rate 1.25 GSa/s (0.8 ns per sample).

The propagation delay was measured using the 50% crossing method. This method finds the time when a signal reaches the middle of its transition from low to high voltage. For the input signal, the voltage changed from 0 V to 3.3 V. The 50% level is calculated as:

$$V_{50\%} = \frac{V_{low} + V_{high}}{2} = \frac{0 + 3.3}{2} = 1.65\ V$$

The input reached this voltage at $t_{in} = -0.41$ ns. For the output signal, the voltage also changed from 0 V to 3.3 V, so the 50% level is the same, 1.65 V. The output reached this voltage at $t_{out} = 7.89$ ns. The propagation delay is the difference between output and input times:

$$t_{pd} = t_{out} - t_{in} = 7.89 - (-0.41) = 8.30\ ns$$

The measured values are summarized in the table below:

Signal	Pre-transition voltage	Post-transition voltage	50% crossing time
Input (C1)	0 V	3.3 V	-0.41 ns
Output (C2)	0 V	3.3 V	7.89 ns

Table 1: Measured values for input and output signals.

From the waveforms, we can see that the output follows the input with a delay of about 8.3 ns. There is a small ringing after the output transition, probably caused by the probe or the circuit, but the signals are clear and stable. Using the 50% crossing method provides a consistent point to measure the delay. Considering the oscilloscope sampling rate and reading accuracy, the uncertainty is approximately ±1.0 ns. The measured propagation delay of 8.30±1.0 ns is reasonable for a circuit with several gates or one driving a load. If only a single fast gate was used, the delay would be smaller, so extra delay may come from loading or measurement conditions.

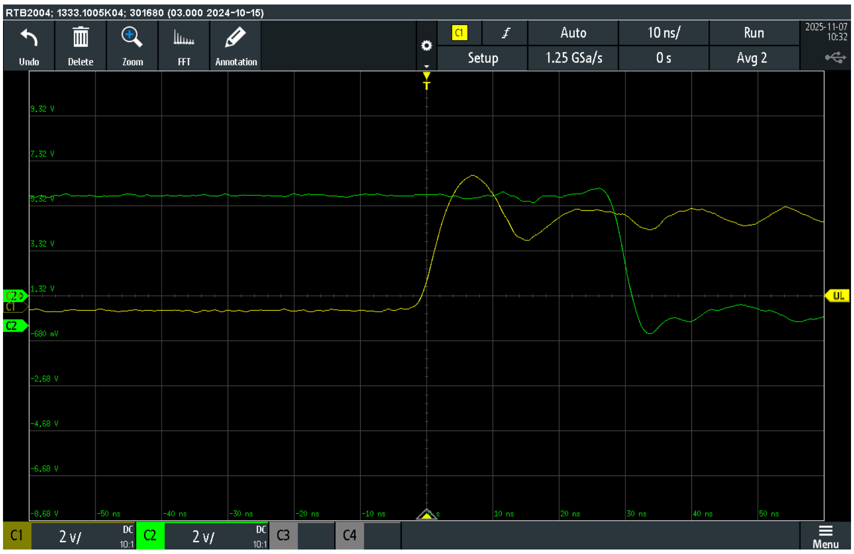

Figure 2: (C1) and output (C2) signals showing 50% crossing points and propagation delay.

3 Task 3 – Glitch Observation

In the lab,we wanted to measure the glitch. However, due to lack of time at the end of the session, we could only partially complete the measurements. Based on the expected behavior, when the input signal changes (rising or falling edge), the output may show a brief unwanted pulse (glitch) before stabilizing. This occurs because different paths in the logic circuit have unequal propagation delays, so some internal signals change slightly earlier than others. The glitch is very short, typically a few nanoseconds, and can differ in size between rising and falling edges. The output may also show overshoot or ringing due to parasitic capacitance and inductance. Any temporary pulse at the output that does not correspond to the intended logic level is considered a glitch, indicating the circuit evaluates the logic incorrectly for a very short time. The glitch appears at the exact moment the input transitions and is not visible when the input is stable. A fast oscilloscope time-base (few ns/div) is needed to observe it clearly. Even though our measurement was limited, the expected behavior is understood: it is a short, unwanted pulse caused by unequal delays and parasitic effects, appearing only during input switching and often more visible on one edge than the other.

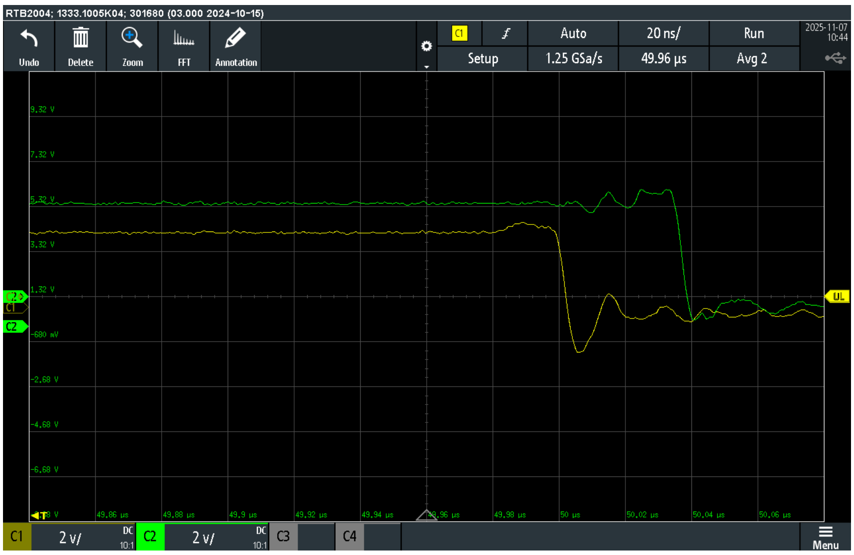

Figure 3: input (C1) and output (C2) signals showing the glitch